

The Application of Latent Semantic Analysis to the Voynich Manuscript

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Abstract

The Voynich Manuscript (VM) is a medieval manuscript likely written in the 15th century (Yale Univ., Beinecke Rare Book & Manuscript Library MS 408).^[1] The manuscript is written in an unknown language or code using an unidentified set of symbols that has yet to be made legible. Additionally, the codex contains many strange and fantastical images of plants, people, and cosmological/zodiac illustrations, the meaning of which are also unknown. One of the main research avenues into the VM is to examine its textual content to understand how it behaves relative to known texts; this can provide insight as to whether the mysterious writings contain decipherable text or not. In this paper, we explore the coherence and flow of the manuscript using Latent Semantic Analysis (LSA). LSA is a technique that may help ascertain whether the behavior of the text within the VM shows evidence of a coherent flow of topical content, by comparative analysis of text samples that are near each other, farther away from each other, at section breaks, or even page breaks. The advantage of this strategy is that LSA analysis can be undertaken without actually knowing the meaning of the text. We expect portions of text that are near to each other to have a relatively high similarity score, that is, to be potentially semantically related. We also expect that at anticipated topic breaks (pages or sections), the similarity score between adjacent text blocks would be smaller, as the breaks seem to represent a change in topic. Both of these patterns are observed in the control manuscript studied as proof-of-concept experiments. Patterns then observed in several sections of the VM suggest that there may be an overall coherence to the text.

1. Introduction

The Voynich Manuscript has been called The World's Most Mysterious Manuscript [Manly 1921], and for good reason. On the one hand, the Voynich Manuscript (VM) is a typical medieval manuscript written and decorated using iron-gall ink and natural pigments, on parchment that has been carbon-dated to the early fifteenth century (ca. 1420). Nothing mysterious about that. But upon opening the codex, the utterly unique features of this intriguing object become immediately apparent. The manuscript is written entirely in an otherwise-unattested set of symbols (known as Voynichese) and illustrated with unidentifiable plants, uninterpretable astrological diagrams, and astonishing biological drawings. The Voynich Manuscript has intrigued, mystified, and frustrated linguists, cryptologists, enthusiasts, and scholars worldwide for centuries. This essay presents a novel approach to the still-unanswered question of whether the manuscript has underlying meaning at all.

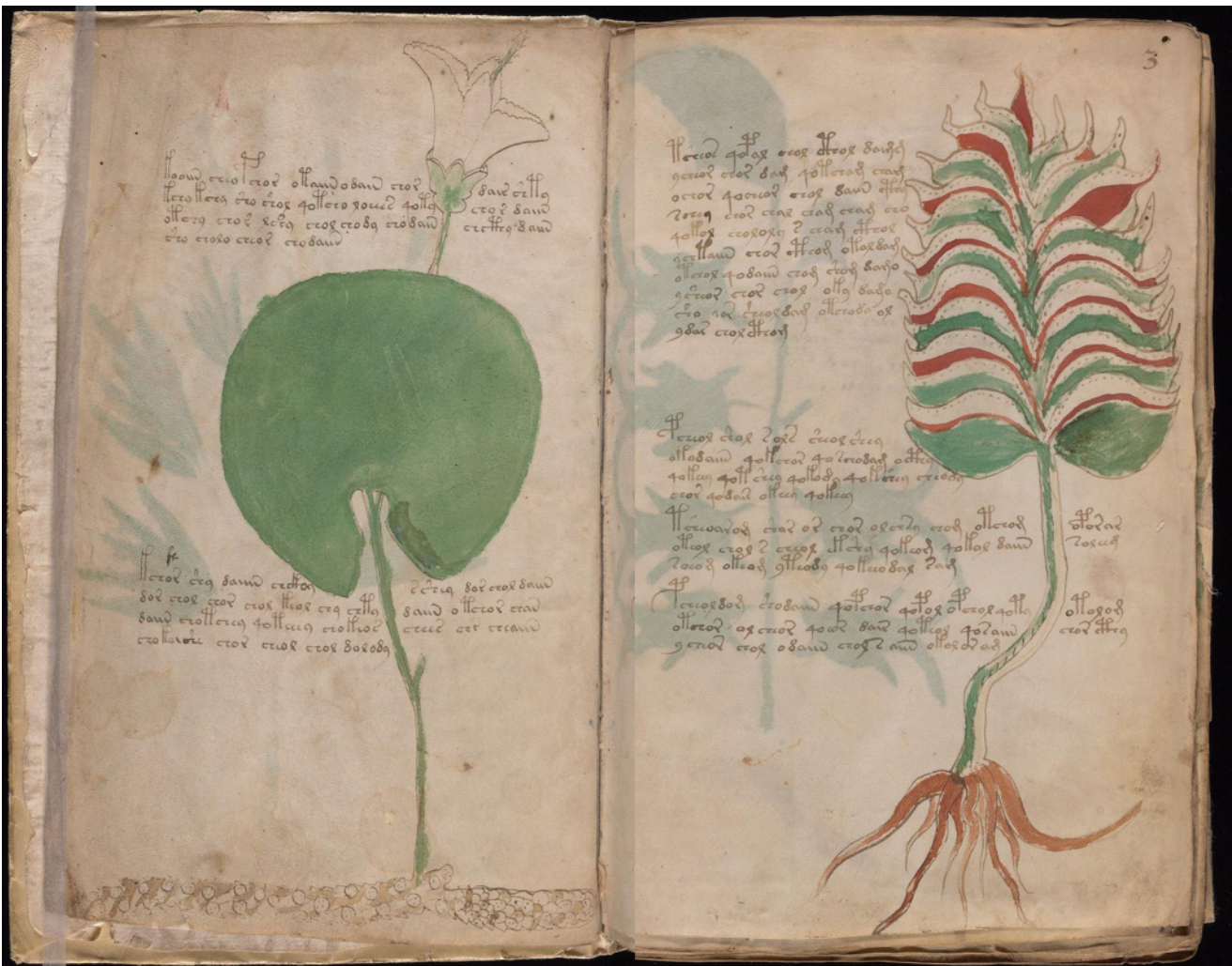


Figure 1. Folios 2v and 3r of the Voynich Manuscript

For the present study, it is important to understand the structure of the manuscript. Examining the manuscript's physical structure reveals that the visual transitions that seem to indicate new topics are also structural transitions. These visual transitions (for example, from a series of illustrations of plants to a series of what appear to be cosmological diagrams) occur at physical breaks between "quires," the gatherings of nested folded sheets of parchment. The illustrative sections, scribes, languages, and structure of the manuscript are summarized in Table 1. In the Quire Structure column, the baseline Arabic numerals represent the sequential numbering of the quires, while the superscript numerals indicate how many leaves are in each quire. For example, " 2^{8-1} " means that Quire 2 originally had eight leaves (four nested bifolia) but is lacking one, and " $3-7^8$ " means that Quires 3 through 7 each have eight leaves. The table below also indicates which of the five scribes identified in [Davis 2020] contributed to each section and which variant type of "Voynichese" is used in each section ("Language A" or "Language B," as first identified and described in [Currier 1976]). It should be noted additionally that Scribe 1 writes in "Language A," while the other four scribes consistently use "Language B." These distinctions will be discussed further below.

Section	Scribe(s)	Languages	Quire Structure
Botanical (ff. 1r-66v)	1, 2, 3, 5	A, B	$1^8, 2^{8-1}$ (lacking f. 12), $3-7^8, 8^{10-6}$ (lacking ff. 59-64)
Cosmological/Astronomical/ Astrological diagrams (ff. 67r-74v)	4	B	$9-11^2, 12^{2-1}$ (lacking f. 74)
Biological (ff. 75r-84v)	2	B	13^{10}
Rose foldout (ff. 85r-86v)	2, 4	B	14^1 (a single large sheet)
Pharmaceutical/Recipes (ff. 87r-102v)	1, 3	A, B	15^4 , [16: lacking], 17^{8-4} (lacking ff. 91-92, 97-98), [18: lacking], 19^4
Stars (ff. 103r-116r) ^[1]	3, 2	B	20^{14-2} (lacking ff. 109-110)

Table 1. Summary of the Voynich Manuscript

In spite of these *visually* coherent sections, even after a century of study we still do not know whether the manuscript demonstrates *textual* coherence. Do the various sections represent a coherent whole, or are they unrelated to one another? Does the text within each section demonstrate internal coherence? Such questions have important implications for “reading” the manuscript. In a textbook, for example, topics discussed in a particular chapter will, by and large, be related to one another and will inherently have some sort of semantic similarity, as a single overarching topic is being addressed. We would expect that the closer two pieces of text are to one another, on average, the higher their similarity, as it is more likely the topics being presented would be closely related. By extension, we expect that the flow or coherence of the text would diminish at a chapter break, as the topic is likely to change. Using techniques such as Latent Semantic Analysis (LSA) we can measure the similarity between two blocks of text (called “documents” below), analyzing relationships between a set of documents and their constituent words (called “terms” below); it is particularly useful for revealing hidden (latent) semantic structures in textual data (rather than semantic meaning) and determining if those relationships hold (for technical details about LSA, see Appendix 1). By applying the LSA technique to known texts (as baselines) and to the unknown text of the Voynich Manuscript, we can address the question of coherence in the VM. If the analysis of the Voynich Manuscript suggests some degree of coherence and flow, this may suggest that there is indeed a coherent message underlying the mysterious text. It is important to keep in mind that we are not making a claim about reading or deciphering the Voynich Manuscript; LSA analytics functions by identifying patterns and context, not semantic meaning.

2. Latent Semantic Analysis

The technique employed in these experiments is known as Latent Semantic Analysis (LSA) [Deerwester et al. 1990], a technique that originated in the field of Information Retrieval (IR). The methodology analyzes sets of documents (paragraphs, articles, emails or other writing) to determine how similar they are to each other. In a given text or texts, it is quite likely that a variety of words will be used to express similar ideas or concepts. For example, there might be a paragraph about cars, one about vehicles, and a third that focuses on automobiles. Human readers know they are related, but a computer will treat the distinct words as separate entities. LSA is a method that identifies these hidden (i.e. “latent”) underlying structures and relationships between the words in these documents by analysing the patterns of

word usage across the compared texts. In essence, because words that appear together in similar contexts tend to have similar meanings, LSA considers which words occur in which documents, finds patterns relating to how these words co-occur with one another, and uses those findings to infer underlying relationships or concepts. Two compared documents that share these relationships/concepts will tend to return a high similarity score.

A full mathematical explanation of LSA can be found in Appendix 1 in addition to a discussion of its strengths/weaknesses and parameter decisions that were made for the present study.^[2] We will provide a brief high-level description here. For a given manuscript, such as the Voynich, we partition it into smaller “documents” approximating a set size (say 20, 30 or 40 terms) after performing some pre-processing on the text. An n by m matrix A (where n corresponds to the number of unique terms and m corresponds to the number of documents) is formed from these documents such that each row is a term, each column corresponds to a document, and each cell in the matrix, say A_{ij} , corresponds to a Term Frequency (TF) count where A_{ij} represents the number of times term i occurs in document j . This representation is known as a *Bag-of-Words* (BoW) model, in which each document is simply a collection of terms and the order of the terms is not taken into account. LSA takes into account which terms occur together (as in the same document) or are used in similar ways and does not use additional positional context of the terms. Further modification of these values may take place in order to scale them to achieve better results.^[3] This A matrix is then transformed using a Singular Value Decomposition (SVD) which, in essence, creates new vectors representing our documents which we can then use in computations to generate similarity scores between them. These new document vectors bring out the underlying “latent” features and relationships between the terms and documents by reducing the noise in the data. We refer to this transformation of A as our “semantic space” representing the collection of processed documents. A is factored into 3 matrices, TSD^T , and it is the columns of the D^T matrix that will be of interest to us (as they represent the transformed space for our “documents”).

In order to compare the similarity of two documents, we calculate the cosine between their vectors (akin to columns in A); the higher the cosine, the greater the similarity. The cosine score is based on the angle between the two vectors. Vectors with a smaller angle between them (generally those that “point” in the same general direction) have a higher cosine and therefore a higher similarity than those that present a larger angle and a smaller cosine. With a known text such as the *Aberdeen Bestiary* (discussed below), we generally interpret this as semantic similarity, because two compared documents, if they have a high similarity score, are likely to be discussing a similar subject or theme. We can confirm this by referring back to the original text. With the Voynich, such referential confirmation is not possible, as we do not know for certain whether or not the Voynich contains semantically meaningful content. In the case of the Voynich, then, this analysis is a measurement of the latent similarity of term usage within and across visually coherent sections. Although we cannot know, until such time as the manuscript is rendered legible, whether these connections are truly semantic or not, we *can* ascertain whether or not the Voynich Manuscript behaves in a similar way to known texts. This may be viewed as evidence that 1) there is in fact semantic content within the manuscript, and 2) visually coherent sections are also textually coherent.

It is worth taking a moment to explain why we believe LSA to be a valid methodology in this case. LSA is a 35-year-old IR technique that has largely fallen out of use in favour of new and more modern Natural Language Processing (NLP) approaches. Because of the unique nature of the Voynich manuscript, however, LSA has great potential. Within the Voynich are only about 35,000 terms. More modern techniques such as Word2Vec [Mikolov et al. 2013] and Transformers [Vaswani et al. 2017] generally rely on document sets in the order of millions or billions (or far more when we consider the cutting edge LLMs like ChatGPT, Gemini, or Grok) in order to generate reliable results. Because the dataset size for the Voynich is much too small for popular modern techniques, we are forced to go back to more “old school” NLP methodologies. LSA has been demonstrated to work well with datasets of this size; for example [Dos Santos and Favero 2015] used a semantic space of only 359 documents to perform automated evaluation of students’ written answers. Latent Semantic Analysis is therefore a natural choice to analyze the coherence and flow of the Voynich Manuscript.

2.1. Previous Related Works

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There are several published experiments that support the choice of LSA for this project. [Foltz, Kintsch, and Landauer 1998] conducted an experiment to demonstrate an approach for predicting the coherence of sets of texts by manipulating textual coherence and assessing readers' comprehension. One takeaway from this work was the creation of an automated methodology that can generate coherence predictions. [Foltz, Kintsch, and Landauer 1998] utilized LSA as a technique for measuring the semantic relationship between two given pieces of text. The first experiment they performed involved four versions of a textbook. A 300-dimension semantic space was created using the first 2,000 characters of each of the 30,473 articles from Grolier's Academic American Encyclopedia (details of this can be found in [Landauer and Dumais 1997]). The metric they used was an analysis of adjacent sentence-by-sentence comparisons using LSA, and a calculation of the average cosine. The original coherence score (i.e. average cosines) was 0.192 and was improved to 0.403 in the final version. Another experiment they performed was to detect discourse segmentation, attempting to identify points in the text where a topic shift occurs (thus enabling the segmentation of the text into distinct topics). The idea behind this is based on the hypothesis that coherence scores should be lower between the portions of the text where the discourse topics change. To test this idea, they used the methodology to predict the breaks between nineteen chapters of an introductory psychology textbook. They created an LSA semantic space for the textbook by partitioning it into paragraphs and created a dataset from which a semantic space of 300 dimensions utilising 4,903 paragraphs by 19,160 unique terms (some paragraphs were removed in pre-processing as they represented references, problem sets, and other components that were not relevant to the discourse). To smooth the predictions, a sliding window of 10 paragraphs was utilized. Two adjacent blocks of ten paragraphs were compared using the cosine operator, after which the window would be advanced by one paragraph. This was repeated over all the paragraphs in the text. The average cosine over the entire text was 0.43 while the average cosine between paragraphs across chapter breaks was only 0.16, a result that was statistically significant. By selecting coherence breaks that were two standard deviations below the mean they successfully identified nine of eighteen chapter transitions. As the cutoff point was increased, more breaks were identified but so, too, were false positives, which increased at a linear rate. Some of the chapter breaks with unexpectedly higher scores (0.25 and 0.29) were found to comprise text that linked the two chapters, accounting for these higher values that are still, it should be noted, well below the mean of 0.41.

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[Timm and Schinner 2023] performed several experiments on the Voynich Manuscript in order to ascertain how related or unrelated the Currier "Language A" and "Language B" variants might be. They utilized a vector space model built from the Voynich Manuscript where documents are created from individual pages. These pages are represented by vectors utilising the Term Frequency (TF) of each term present on the page (so the vectors will consist mostly of zeroes — this is analogous to the columns in the A matrix described earlier). They concluded there is no sudden transition between the two "languages", A and B, and that there is a gradual evolution between them after experimenting with cosine calculations between all possible pairs of pages. One possible drawback to the Timm & Schinner approach is their reliance on the TF model of the text. A cosine will only score a non-zero value if there is text overlap between the two documents being compared. While valuable, the TF approach does not identify underlying latent relationships, structure, or synonymy. By contrast, LSA has the advantage of recognising and scoring relationships between words, identifying relationships between non-identical words that may be functioning synonymously. Additionally, TF values can skew results because terms that are very common can artificially inflate the similarity scores. Weighting methods such as Term Frequency-Inverse Document Frequency (TF-IDF) and Log-Entropy^[4] can help smooth out these discrepancies.

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[Reddy and Knight 2011] performed a similar page-wise comparison experiment on the Voynich Manuscript that focused on whether or not it was likely the leaves of the manuscript are in the correct order. Like [Timm and Schinner 2023], they created a TF vector for each page and performed page-wise comparisons using the cosine operator, results which suffer from the same weaknesses outlined above. They published results of page comparisons with the pages "scrambled" in which the scrambled adjacent-page similarity scores were close to 0, while they showed higher similarity when pages were scored in accordance with the manuscript's current page sequence. They do, however, note that some of the non-contiguity of the herbal/pharmaceutical sections and the A/B language distinctions may suggest that some pages were probably re-ordered.

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[Sterneck, Polish, and Bower 2021] investigated topic modelling in the Voynich Manuscript using techniques such as

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Latent Dirichlet Allocation (LDA), Latent Semantic Analysis (LSA) and Nonnegative Matrix Factorization (NMF). The objective was to try and cluster the text in the Voynich to look for any interesting clusters of subjects or topics. This included using features such as Lisa Fagin Davis's five scribal corpora, A/B language classification, visual subject label (astrology, botanical, balneological, rosette, recipes/starred section and unknown), and individual quires. The text was vectorized on a page by page basis (with TF-IDF weighting applied to the NMF/LSA methods to increase the visibility of rare words and decrease the influence of words that are quite common) and fed into the cluster algorithms after being transformed by one of the previously mentioned methods (LDA/LSA/NMF) and introducing a dimensionality reduction phase (reducing the high-dimensional space created by these methods into a lower-dimensional space that may enable topics or concepts to be more visible in the analysis). The k-means clustering algorithm was then utilised and results presented. Initial results were generated using all three methods, but additional experiments were continued using only NMF; this was due to identified issues with LDA (picking optimal parameters) and potential issues with LSA, a technique that does not perform well with polysemy or homophony as such relationships dilute the method's ability to discover underlying latent connections between word usage. Entropy experiments in [Lindemann and Bown 2021] suggest this may be an issue, although our experiments either did not encounter these issues or found them to be negligible, if present at all. Based on their experiments (largely using NMF), Lindemann and Bown suggest the text in the Voynich Manuscript likely represents an enciphered human language rather than "meaningless" text and that the illustrative topics can be clustered to some degree.

3. Methodology

The framework of the experiments conducted here is based on the research by [Foltz, Kintsch, and Landauer 1998] which was briefly highlighted in the previous section. Specifically, we will use their method for determining document coherence as well as performing discourse segmentation on the Voynich Manuscript and a select known text.

The coherence experiment is fairly straightforward in principle. Given a semantic space for a particular text, we consider all the adjacent columns in the D^T matrix. The columns, from 1 to m (where m is the number of documents) are in the same order in which they were processed from the original text. That is, column j and $j + 1$ are adjacent in the original text. To perform a coherence measurement, we simply take the cosine scores between all the adjacent documents from 1 to m the pairs (1, 2), (2, 3), ..., (j , $j + 1$), ..., ($m - 1$, m). The average of these scores is then calculated, and we treat that as the coherence score. A second set of calculations is performed by taking a permutation of the order of the documents that are compared such that no two adjacent documents are present in the permutation and comparison scores (and the average) is once again calculated. This is done 10 times for the random permutations, and an overall average is calculated. The two values can then be compared. It is expected that the average for the correct-order documents is significantly higher than the average for random permutations.

For each dataset used in the coherence experiment, a number of semantic spaces will be created. These will use documents of approximately 20, 30 and 40 terms, each which will be represented in the semantic space as individual columns in D^T . Additionally, before the creation of the semantic space via the SVD operation, the A matrices will be weighted using TF, TF-IDF and Log-Entropy weightings. Each text will therefore have a total of nine semantic spaces generated for the experiment to allow for comparison between them. The comparison mechanism (using a document of, say, size 30) for the coherence scores can be seen conceptually in Figure 2. The similarity score is the cosine value of the two vectors representing the documents from the semantic space.

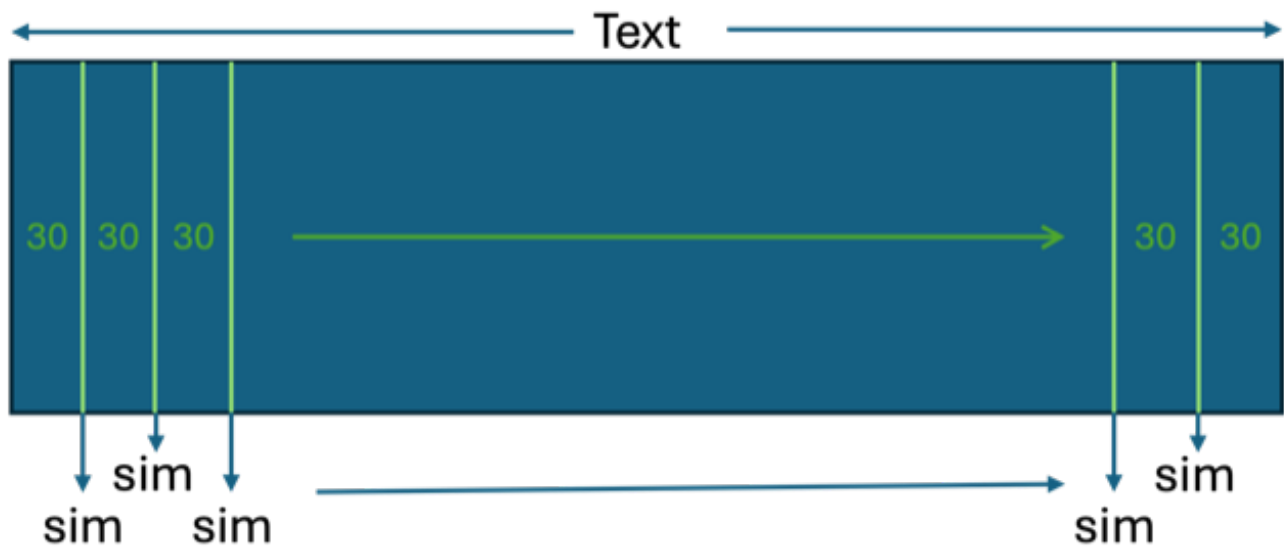


Figure 2. Schematic Representation of Size 30 Windows

With the discourse segmentation experiment, we can test the ability of the methodology to identify significant features such as section breaks, chapter breaks, page breaks, or other transitions. This experiment uses the Voynich Manuscript and one known text (the Latin *Speculum Humanae Salvationis* ^[5]) so we can compare the behaviours between them. If we can detect semantically-significant features in the known text, then it may indicate that similar features or behaviors in the Voynich suggest some sort of semantic structure.

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For this experiment, the data sets were partitioned into logical sections: For the *Speculum* we used chapters and for the Voynich we used pages. Within each Voynich page, paragraph, circular text, and radial text were grouped separately for further partitioning into documents (because we do not know whether these different formats serve different functions within the manuscript). We then generated documents from these partitions of approximately size 20 and size 40^[6]. The size 40 documents, utilizing TF-IDF weighting as per the previous experiment, were used for creation of the LSA semantic space; the size 20 documents were utilized in the discourse segmentation experiment explained below.

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The documents created for the discourse segmentation comparisons were windows of 10 concatenated documents from those generated of size 20 (for VM pages with sparse text, the concatenated “documents” were obliged to span more than one page). These concatenated documents were used for side-by-side comparisons in sliding windows. The larger concatenated documents functioned to reduce the variance between comparisons and create a smoother picture of the flow of the discourse, as well as detecting when changes may be occurring.

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The moving window of textblocks is illustrated in Figure 3, showing the comparison of two concatenated documents (of 10 x 20 each). The next comparison is made by sliding each of these concatenated documents over by one document, resulting in the introduction of new text (and overlap with the previous comparison). This is repeated until we have exhausted all the text windows.

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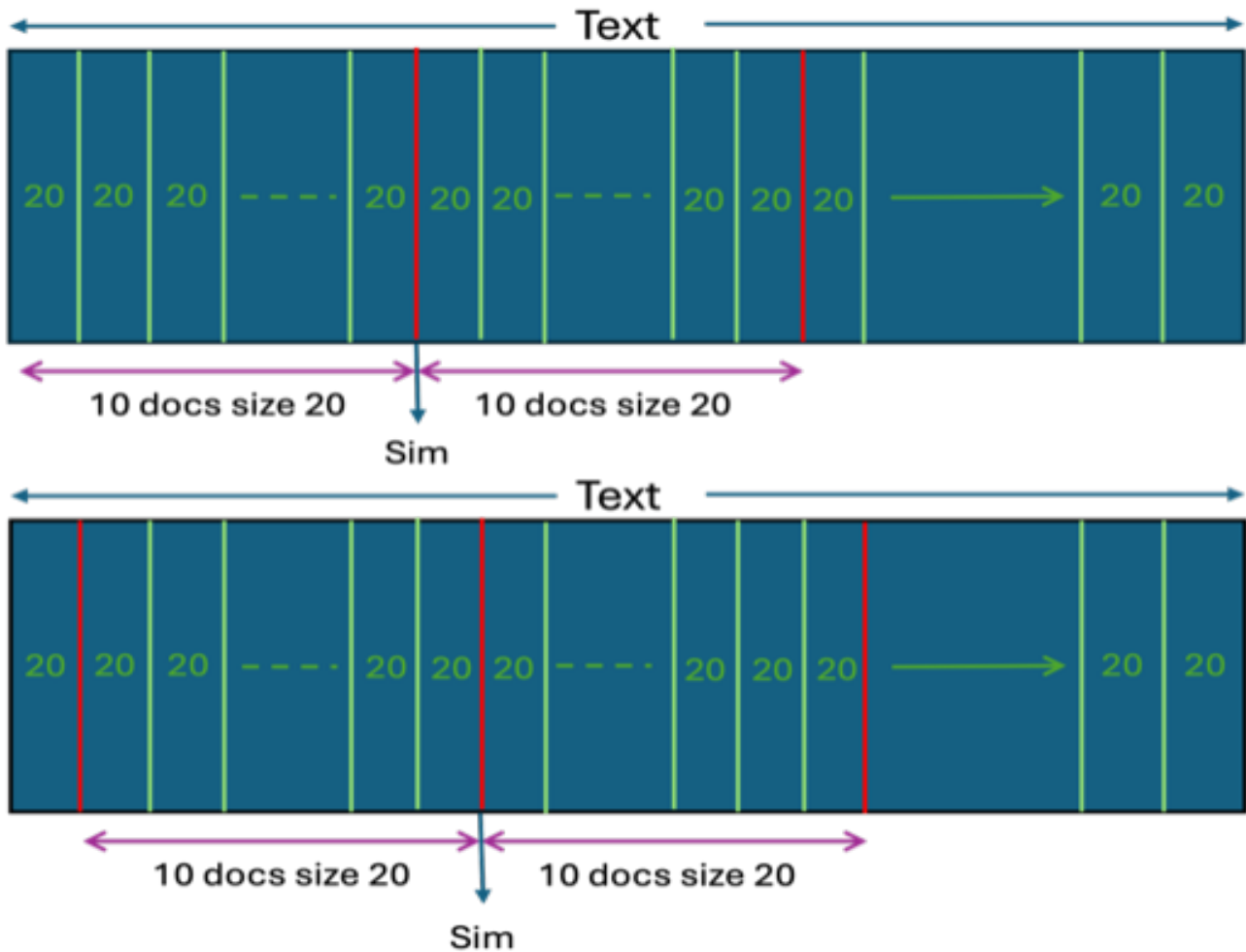


Figure 3. Discourse Segmentation Windows

An example of this step is given below. For the Voynich the first two 10 X 20 documents are represented as:

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1r-1,1r-2,1r-3,1r-4,1r-5,1r-6,1r-7,1v-1,1v-2,1v-3[—] 2r-1,2r-2,2r-3,2r-4,2v-1,2v-2,3r-1,3r-2,3r-3,3r-4

In this example, 1r-1 is the first document of 20 terms on f. 1r, 1r-2 is the second document of 20 on f. 1r, and so on. The window moves to 1v-1 after all of the 1r text has been segmented. 1v-3 is the tenth and final document of this window. The symbol [—] represents the transition from one group of ten to the next. The concatenated document to be compared begins with 2r-1 and proceeds by groups of 20 until the tenth, which is the fourth document on f. 3r. The next comparison would shift the window by one segment, beginning with 1r-2 and ending with 3v-1:

1r-2,1r-3,1r-4,1r-5,1r-6,1r-7,1v-1,1v-2,1v-3,2r-1[—] 2r-2,2r-3,2r-4,2v-1,2v-2,3r-1,3r-2,3r-3,3r-4,3v-1

The process continues until the last window advances to the end of the text. Each similarity score is recorded, paying particular attention to page, section and scribal changes.

3.1 Datasets

For the initial coherence experiments, we employed several open-access texts in addition to the VM. These texts were chosen because they represent a variety of linguistic and stylistic genres: George Orwell's *Animal Farm* (approachable contemporary English prose in a coherent narrative); The *Aberdeen Bestiary* (didactic and formal medieval Latin and a modern English translation, with each chapter representing a different and clearly-defined topic)^[7]; Dante's *Inferno* (rich and poetic medieval Italian in a chapter-based narrative); the *Speculum Humanae Salvationis* (didactic and formal

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medieval Latin in topically-varied chapters), and computer-readable Voynichese generated via the software^[8] kindly provided by Torsten Timm [Timm and Schinner 2020]^[9]. For the VM ASCII transcription, we used ZLversion 2a dated 17/03/2022, which is the most complete, consistent, and accurate transcription available.^[10] The wide variety of chosen texts should prove useful to demonstrate the coherence measurement and the behaviour of the texts in these contexts. We would expect them all to behave in a somewhat similar fashion regardless of the style of the writing. The artificial text generated via the Tim and Schinner (2020) algorithm is meant to mimic the Voynich Manuscript text, according to their self-citation hoax hypothesis, and preserves some of its statistical features (for example, the generated text should follow Zipf's law). Their hoax hypothesis states that the text gradually changes over time based on various observations they have made on the manuscript (that many words in close proximity with one another are also very similar to one another — for example Voynich terms such as *qokeedy* and *qokeey*. Although we can't use this to generate every section of the Voynich Manuscript individually, we used it to create a set of text roughly the same size as the Voynich. The algorithm only simulates a subset of the manuscript text and there is no language A/B distinction (they do not subscribe to the fact that there is anything really significant between them and it is a natural function of this gradual shift in text/words over time that causes it); in their experiments they generated a portion of text of roughly the same size as recipes section in the manuscript.

Before we can utilize LSA, there are several steps that must be undertaken to prepare each text for further use. These steps are typical for the application of many NLP algorithms, such as lowercasing or removing punctuation and stop words. As the VM cannot be "lowercased" (since it has no identifiable upper- or lower-case characters), we pre-processed all of the baseline texts in other ways for consistency: removing punctuation and using an algorithm to tentatively identify and remove the stop words. Stop words are common terms that, due to their grammatical function and frequency, offer very little value to the semantic discourse. In practice, stop words act as background noise that can dilute other relationships in the text. In English these are words such as "the", "of", "is", etc. Again, in order to treat all of the texts consistently, stop words were determined for all texts using an automated method, detailed in Appendix 1. Because we do not know what the stop words are in the Voynich Manuscript, we used the same algorithm for all of the texts instead of referring to pre-existent lists of stop words in the relevant languages. This gives us a like-for-like basis of comparison.

The VM ZL Transliteration file underwent additional pre-processing. The file follows the Intermediate Voynich MS Transliteration File Format (IVTFF) version 1.7 as defined in (Zandbergen, IVTFF - Intermediate Voynich MS Transliteration File Format - File Format 1.7 2020) that categorizes the text present in the VM. Voynichese "words" (terms) are generally classified based on their graphic context: In the IVTFF transliteration, each term is tagged as occurring in a standard paragraph, standing alone as a "label," or written in a non-standard layout such as around a circle or radiating from a central hub. Our analyses typically disregard "labels" as they offer little to the discourse and more than 50% of them are *hapax legomena* (terms that only occur once in the text). "Paragraph" texts behave in ways that are similar to the legible texts in our comparison datasets and thus are most useful for LSA analytics. We also removed any terms from the VM transliteration that are tagged as uncertain in any way, such as terms with unclear spacing or ambiguous characters.

Table 2 presents statistics for the processed texts, including total word count, size of stop word list, word count after stop-word removal/pre-processing and the total number of *hapax legomena* (unique words) in the text. Note that it is the total terms after stop word removal that are used for document segmentation as well as construction of the semantic space although in practice hapax terms are not considered in similarity comparisons scores, as they only occur once and the SVD operation therefore has no context for comparison.

Text	Language	Total Words	Stop Words (SW)	Total After SW Removal	Hapax Legomena
Animal Farm	English	30,023	31	18,360	2,228
Aberdeen Bestiary	English	71,001	47	38,886	4,221
Aberdeen Bestiary	Latin	45,109	17	37,236	9,407
Dante's Inferno	Medieval Italian	32,383	18	24,164	4,642
Speculum Humanae Salvationis (Vincent of Beauvais)	Latin	40,224	17	33,773	6,677
Voynich Generated	Artificial	31,950	27	19,636	1,265
Voynich – ZL	?	31,955	25	24,324	4,802

Table 2.

Table 2: Post-Processing Text Statistics for Coherence Test Data

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4. Results

The results for both the coherence and the discourse segmentation experiments are reported below, followed by discussion and analysis of these results.

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4.1 Coherence Experiment

For each LSA semantic space, we compared the first document (of size 20, 30 or 40) to its adjacent document in order to calculate a similarity score; we repeated the experiment by shuffling the sequence of compared documents such that no two consecutive documents were compared. Since shuffling them is a stochastic process, this was repeated 10 times, and an average of the ten scores was calculated. In other words, we ran the two tests on each of the seven textual datasets: one by calculating the average similarity of adjacent documents, and another using a shuffled set of documents. For each of the text samples, nine semantic spaces were created by applying three weighting methodologies (TF, TF-IDF, Log-Entropy) to three different document sizes (20, 30, 40). The results (i.e. the average similarity score across the entire text) are reported in Figures 4-6:

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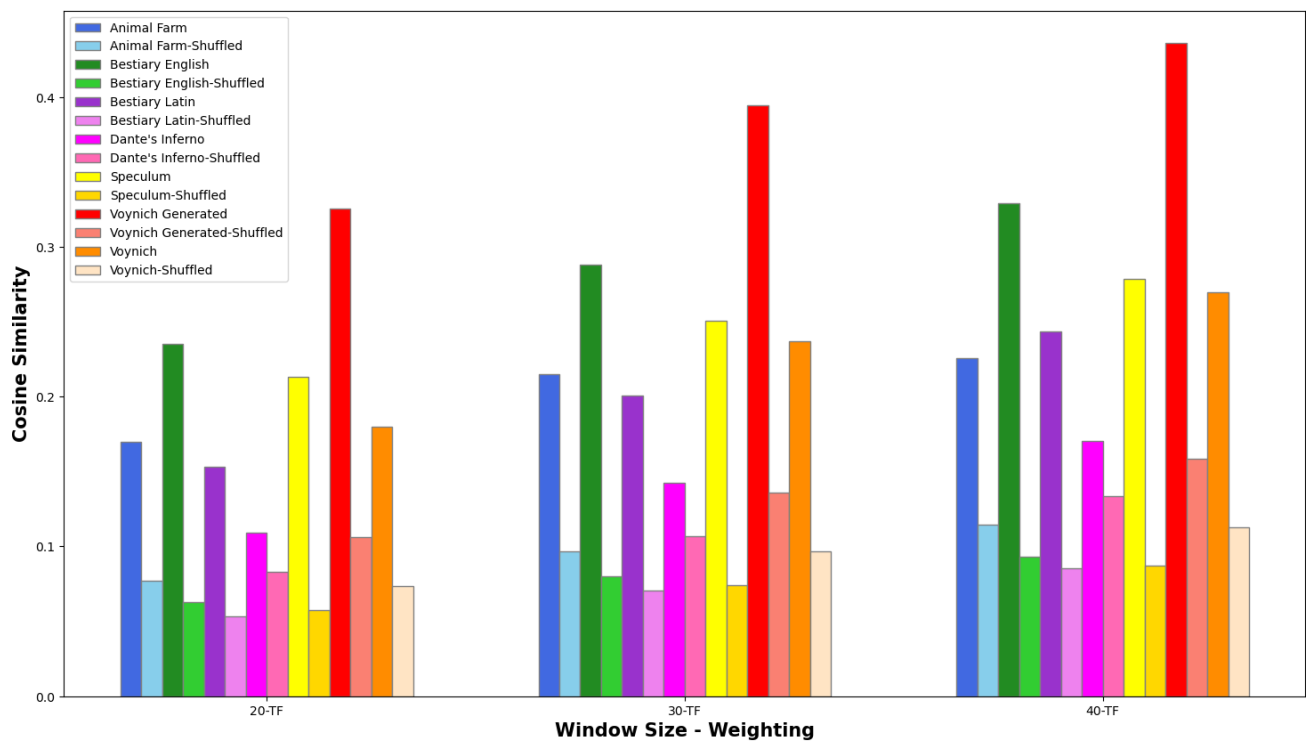


Figure 4. Coherence Test Results using Term Frequency (TF) Weighting

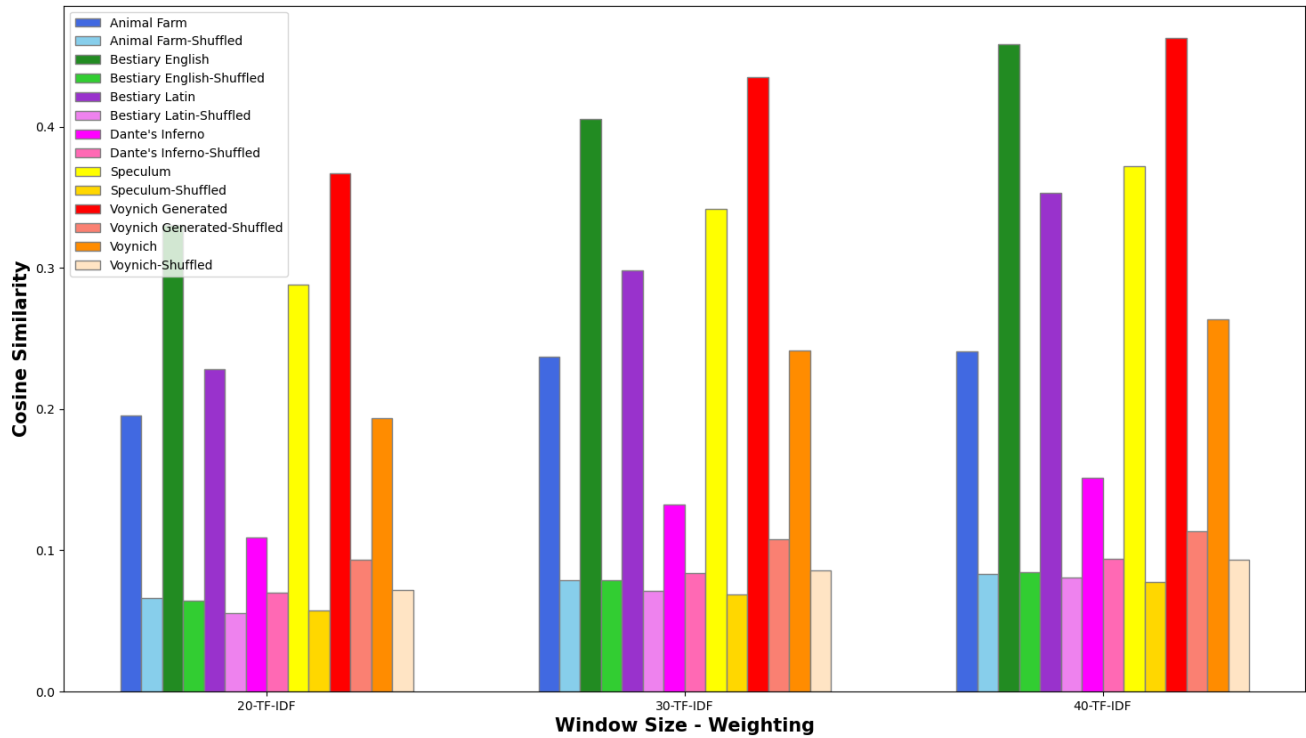


Figure 5. Coherence Test Results using Term Frequency – Inverse Document Frequency (TF-IDF) Weighting

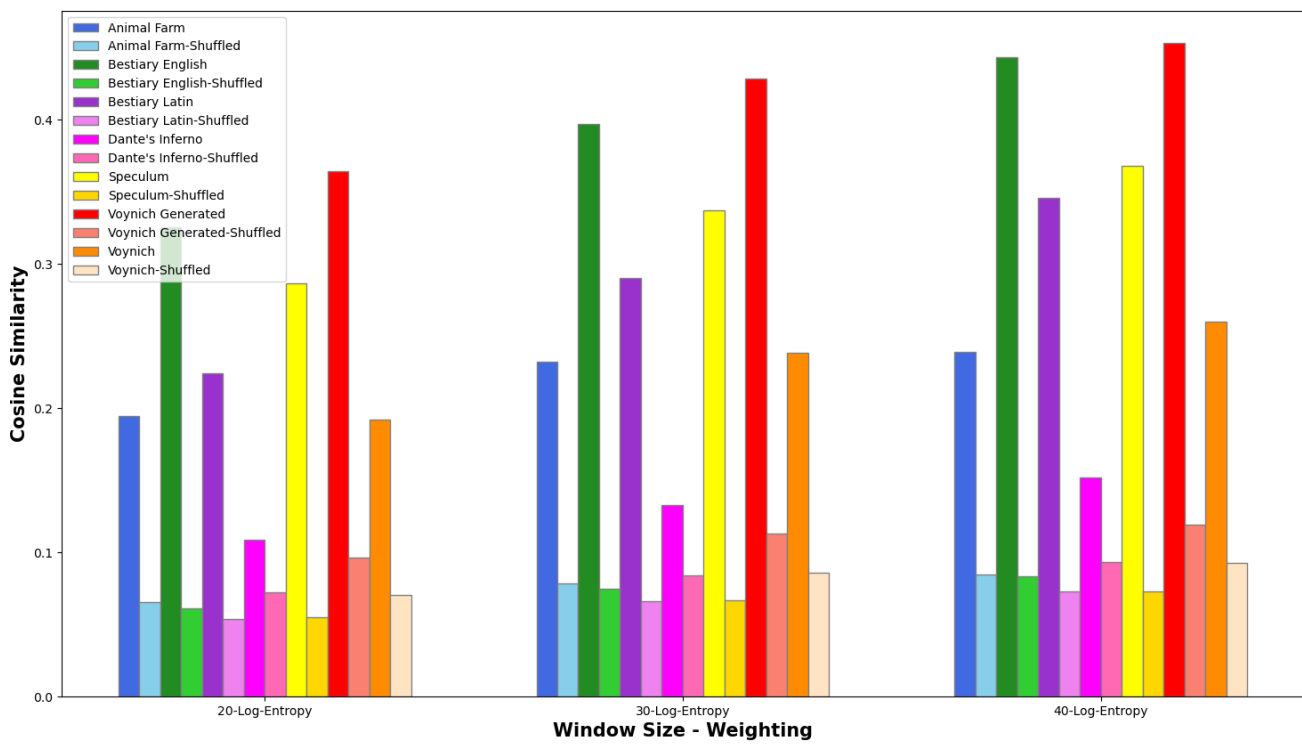


Figure 6. Coherence Test Results using Log-Entropy Weighting

In each case, the average similarity for the in-sequence text was significantly higher than the shuffled-text results, demonstrating that the method correctly responds to adjacency vs. non-adjacency. The TF weighting (Figure 4) gave the lowest scores, while TF-IDF and Log-Entropy produce very similar results. The LSA semantic space of 40 resulted in the most visible contrast between “correct” and shuffled texts. For this reason, we used documents of (approximately) size 40 using TF-IDF for the discourse segmentation component of our experiments. The charts demonstrate that the results on known texts are in line with what would be expected.

Additional statistical tests were performed to determine if the results were statistically significant. The Shapiro-Wilk test and D’Agostino’s K0 squared test were performed on the cosine values to determine if the raw similarity score distribution was normal; this was negative in all cases. We performed the Mann-Whitney U to test for significance. All of the comparisons scored well below a p value of 0.05 — in fact most were close to zero with the highest p value recorded being 0.000145. In other words, the results were indeed statistically significant.

4.2 Discourse Segmentation Experiment

The experiments above demonstrated that the text in the Voynich Manuscript behaves as would be expected for a meaningful text. In the following experiments, we will test the ability of a methodology called “discourse segmentation” to identify significant features such as section breaks, chapter breaks, page breaks, or other transitions, using the *Speculum* as a proof-of-concept text. A successful experiment would indicate that the same method might be able to detect similar features in the Voynich Manuscript.

We first applied this to the known text, the *Speculum*. This was done on two versions of the *Speculum*. We used both the original *Speculum* and a ‘cleaned’ version (removing clauses that do not contribute to the semantic reading of the text such as the words “Chapter 1” or the summaries of the previous chapter that begin each new one). This is in line with the work done by [Foltz, Kintsch, and Landauer 1998], the source of this technique. For the unaltered *Speculum* dataset, the average similarity score across the entire text was 0.440 and the average score at chapter breaks was significantly lower, 0.225. This shows a clear distinction between the complete text and the chapter breaks overall. Using the cleaned version of the *Speculum*, the results show slightly more contrast, with a higher overall average of 0.445 and a lower average score at chapter breaks of .212. This indicates that removing the small amount of redundant

text at the beginning of each chapter allowed for a more accurate comparison of the actual content of each chapter. It should be noted at this point that all further results reported using the *Speculum* dataset use this ‘cleaned’ version.

We used the bottom 5% of the results as a cutoff point for investigation.^[11] Can characteristics of the compared text samples explain the low scores? Are the compared documents truly dissimilar, or are they false negatives? In these explanations, we will reference a generic concatenated document of size twenty as d1-d2-...-d9-d10 [—] d11-d12 ...-d19-d20, where “dx” is a document, and [—] represents the transition from one cluster of ten to the compared group.

For the cleaned *Speculum* experiment, the bottom 5% are represented by 82 scores, with a cosine cutoff point of approximately 0.128. The transition between the two groups of ten occurs at a chapter break in 12% (10) of these low scores, identifying a change of topic and explaining the low score. This value rises to 49% if we include scores where the chapter break occurs not at the [—] transition but at documents that are near it, ranging from d7 to d14. Reference to the text of the *Speculum* (reading the Latin) identifies major topic changes or unrelated content in 51% of the lowest scores. Figure 7 visualizes these results. Each dot represents the cosine comparison score between two concatenated documents of 10 X 20, presented in order sequentially from left to right. The vertical red lines represent chapter divisions, and the horizontal green line represents the lowest-5% cutoff. Upon reference to the Latin text, it becomes clear that the high scores do indeed represent text samples that are related in content. For example, the cluster of high scores in the upper right corner represent chapters that record prayers directed to and describing events in the life of the Blessed Virgin Mary. The sequence of low scores at the far right reflects the transition of the text into a summary conclusion that leaps from topic to topic.

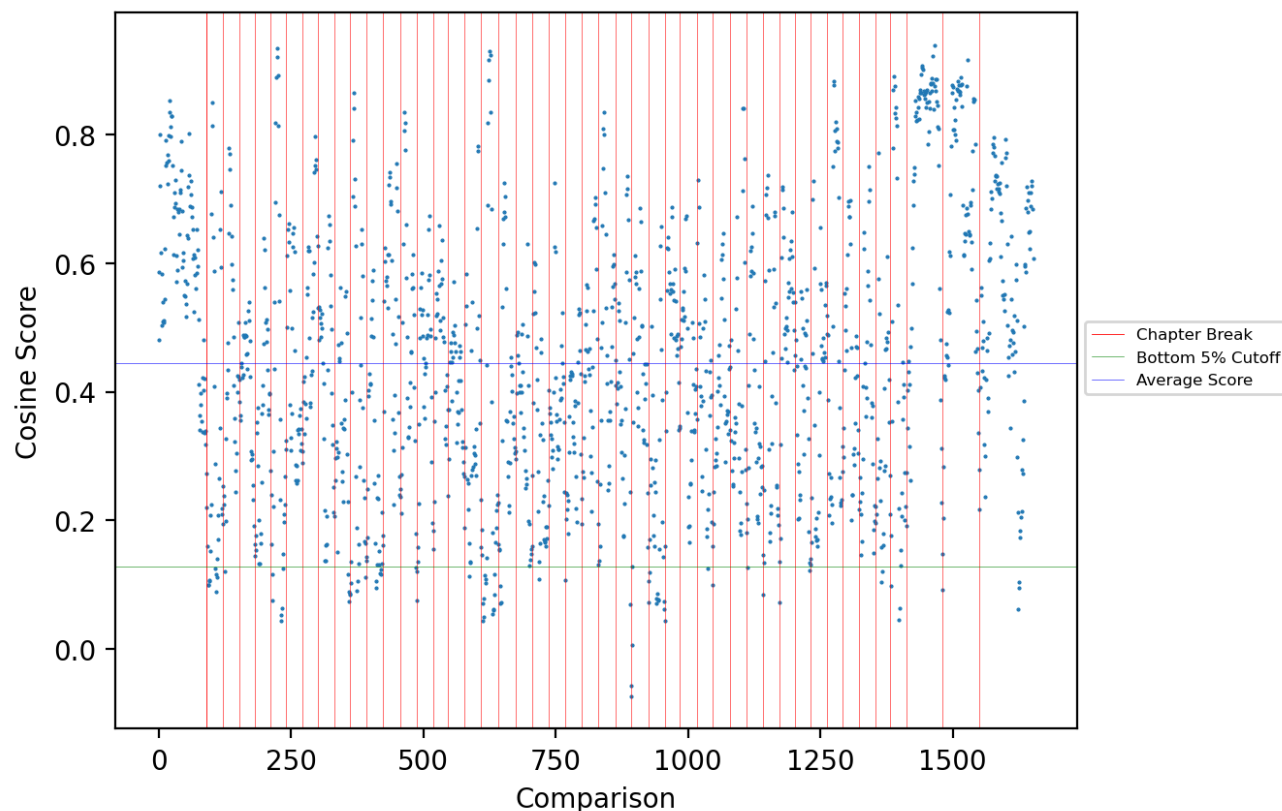


Figure 7. Discourse Segmentation Results Cleaned-Speculum

Having established that the concatenation experiment successfully identified similar and dis-similar text in addition to being sensitive to chapter changes, we ran the same experiment on the Voynich, with intriguing results. Overall, the similarity average was 0.483 while the average similarity over page breaks was found to be 0.421. The average section break score (for example at the Zodiac → Biological transition) was 0.323; this lower-than-average score suggests that those sections could indeed be concerned with different subjects, as suggested by the differing illustrative content, and that the semantic space could be identifying topic changes between sections.

In addition to section breaks, we also examined the scores at scribal transitions. These average scores were even lower, only 0.232. Most scribal transitions represent language (A/B) or section breaks, which explains this result. As described above, Scribe 1 writes entirely in Language A while Scribes 2, 3, 4, and 5 use Language B. Upon closer inspection of the scores at scribal breaks, we observed that this method is particularly sensitive towards scribal changes involving Scribe 1, that is, at transitions from Language A to Language B. We had anticipated that the analysis might reveal this distinction, although the magnitude of the distinction was surprising. The average similarity score at A→B transitions (that is, transitions from Scribe 1 to any other scribe and vice versa) is only 0.206, while B→B scribal changes had a higher average score of 0.332, a substantial difference (there are no A→A scribal transitions because Scribe 1 is the sole user of Language A). This suggests that there is indeed a substantive and significant difference between Language A and Language B.

Conducting the same bottom-5% analysis for the Voynich (with a cosine similarity score cutoff point of about 0.205), we found that most of the lowest scores can indeed be explained by reference to the manuscript itself. Of these approximately-sixty lowest scores, 20% (12) were A→B transitions, and 6% were section transitions. The overwhelming majority of the remaining scores (77% or 46) were Herbal → Herbal page comparisons, virtually all of which were near or on a page break.

As with the *Speculum*, a visualization of the Voynich results is provided in Figure 8, with color indicating the different sections and red vertical lines indicating A ↔ B transitions. The horizontal green line represents the lowest-5% cutoff. Upon closer inspection, the numerous A→B scribal changes in the Herbal section (the tight cluster of red lines) are reflected in the data, as the scores visibly decrease and cluster in the bottom half of the gray section. Also of interest are the higher scores in the Biological (orange), Pharmaceutical (salmon), Zodiac (dark green), and Stars (green) sections — very few of which are below the 5% line — indicating that these sections may concern coherent and distinct topics.

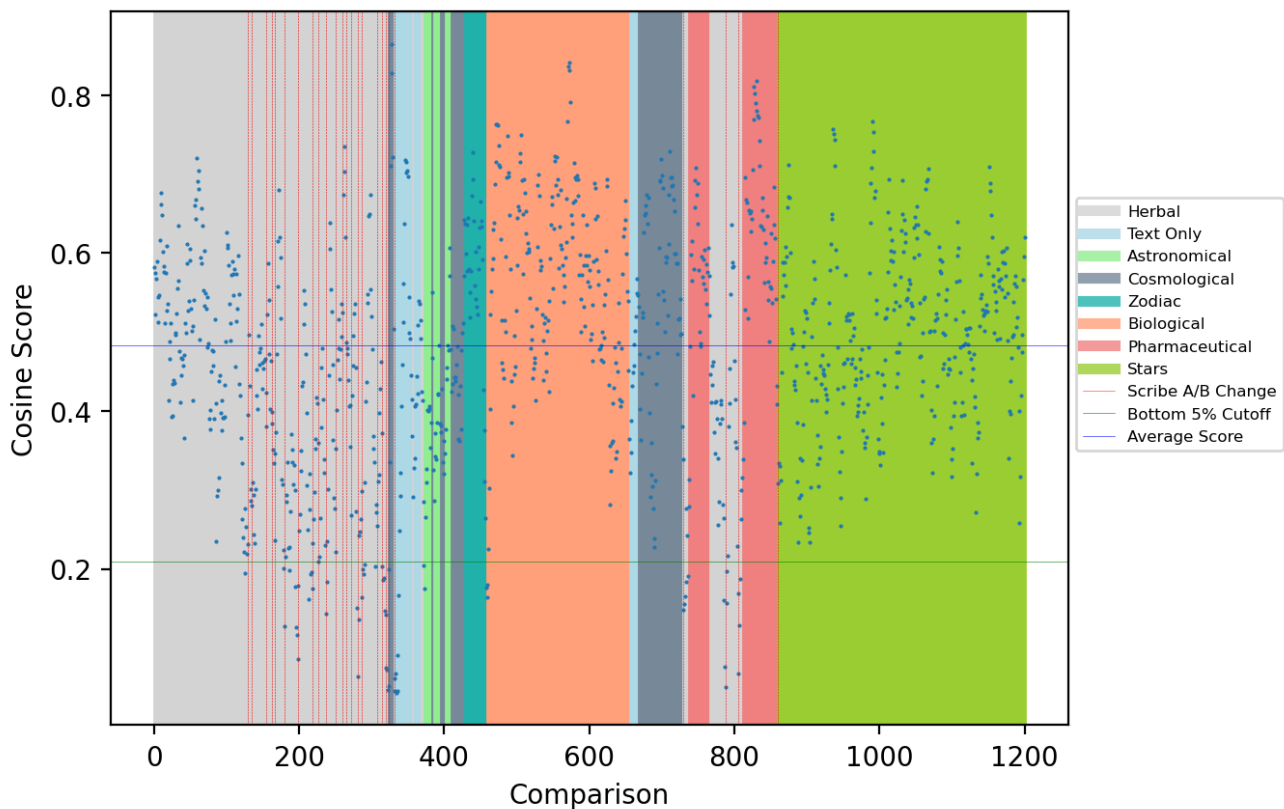


Figure 8. Voynich Manuscript Discourse Segmentation Results

5. Discussion

The coherence experiment was largely in line with our expectations. Our results demonstrate that the average cosine values of correctly-ordered selections of text are much higher than the values for the shuffled text, with a difference that is statistically significant. This reflects the discourse coherence in the baseline, known texts, but also suggests by extension that the Voynich Manuscript (the dark orange and light orange columns in Figures 4-6) does indeed demonstrate coherence across the illustrative sections in a way that is similar to the other known texts; in other words, the proximity of words is indeed significant. This result speaks to the overarching question of whether the manuscript presents a semantically-interpretable text or is random nonsense. The preponderance of evidence suggests that the manuscript may contain semantically-interpretable text.

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There are a few outliers in the results, however. For example, the results for Dante's *Inferno*, although statistically significant, are less dramatic than the other texts present. One possible explanation for this observation may be found in the writing style, as Dante's *Inferno* is a poem with specific word and phrase choices that may impact the results in unexpected ways. The two English examples, *Animal Farm* and the *Aberdeen Bestiary*, also seem to have fairly different values while the *Aberdeen Bestiary* scores quite high in the ordered coherence score. Again, the explanation likely has more to do with the style of the writing in the texts, *Animal Farm* being a novel while the *Aberdeen Bestiary* is a catalogue of animals and fantastical creatures; the two texts have very different subject matter and style of delivery. The deeper investigations that might further explain these particular results are outside the scope of this study.

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Another finding that requires explanation is that the faux-Voynich generated text showed a very similar pattern to the Voynich Manuscript results in that they have the same relative "shape," although the magnitude of the generated Voynich scores was larger (see Figure 5 above). It is also interesting to note the difference in hapax values between the two as well as the difference in stop words removed; however, this will be a function of how the text is generated. The generation of artificial text, roughly, looks at text nearby and potentially modifies some of the words to include as generated text as it proceeds; there is a stochastic element to this. As the artificial Voynich text follows the "self-citation" hypothesis put forward by [Timm and Schinner 2023], this might have been viewed as evidence in favour of concluding that the manuscript has no semantic content; however, this is offset somewhat by our findings regarding transitions from Language A to Language B (see Figure 8 above). Their explanation of the self-citation hypothesis states that the difference between Language A and B is a natural side effect of the self-citation/modification of the text by the scribe over time. The startling difference between A/B in our experiments, however, show this change to be very abrupt and occurs mostly in the same section (Herbal) which would seem an odd place for this to occur.^[12] Our experiments showed the opposite in the sensitivity to the transitions between A and B, suggesting there *is* a real difference between them. It is also worthwhile to note the discourse segmentation scores suggest that some sections contain a higher sense of internal coherence than others (which makes sense, given the illustrative content) — if text were truly artificially generated using a method similar to what [Timm and Schinner 2020] proposed, this would seem to be an odd behaviour as we would expect a more consistent sense of coherence going through the text, not what we are witnessing with the Voynich. The statistical skewness and kurtosis values of the Voynich raw cosine scores (with TF-IDF 40 values for example) are very different than those calculated for the artificially generated values too suggesting a different underlying pattern and distribution (0.48/0.44 vs 0.06/-0.63). The Voynich, for example, has circular and radial text on some of its pages and the algorithm used to create artificial Voynich does not try to simulate things such as this. However, a scribe using such a system would have freedom as to how and when to "make changes."

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The discourse segmentation results were enlightening (see Figure 7 above). For the *Speculum*, we see there is generally a drop in scores at the Chapter breaks, which is to be expected and falls in line with the results of [Foltz, Kintsch, and Landauer 1998] in their experiments they recorded. We could also explain the cluster of high scores in the upper right of Figure 7 as well as the sequence of increasingly-low scores at the end of the text by referring back to the text itself. The overall behavior would seem to confirm LSA is working as expected. In the Voynich Manuscript we see that the Herbal section appears to have some minor coherence at the beginning, coherence that deteriorates quickly once the text transitions from and between language A and B. This LSA sensitivity to the difference between A and B was startling. This observation has implications for the LSA semantic space, as identical words that may be used in a substantially different way in A and B pages could suppress the scores in comparisons that include documents in both Language A and B; this may be a topic for future experiments.

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1. Herbal section: Beginning with the first A→B transition, the scores show a marked decrease, suggesting that the A→B transition is significant.
2. Cosmological, Astronomical, and Astrological sections: The documents in this section (written entirely in Language B) behave similarly to the Language A section of the Herbal portion of the manuscript, that is, they record slightly lower than average scores at page transitions. As each diagram in these sections are unique (and uninterpreted as of yet), we can conclude from this analysis that each page is likely a different topic. In the astrological section, this contention is supported by the images, which focus on a different zodiac sign on each page.
3. Biological, Pharmaceutical, and Stars sections: The results in these sections seem to clearly indicate a marked textual cohesion in each of these three sections, as they show mostly above-average scores with very few scores in the lowest 5%. This reflects the internal cohesion suggested by their illustrative content (for the Biological and Pharmaceutical sections) and common format (for the Stars section).

6. Conclusions

The application of Latent Semantic Analysis to the text of the Voynich Manuscript has proven to be a productive and novel method of analyzing the contents of the manuscript. We have demonstrated the effectiveness of the method by applying it to several known texts and manuscripts, established the general quality of coherence and flow across the Voynich Manuscript, and uncovered previously unseen and surprising features of the Voynich text.

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In particular, we have demonstrated that the Voynich Manuscript seems to follow a coherence pattern similar to known texts. That is, text samples that are closer together, in general, have a higher “similarity score” than text samples that are farther apart. The difference between the known texts and Voynich text is that with known text we can refer back to the text to explain the results — high similarity scores do indeed generally reflect topical similarity. With the Voynich Manuscript we cannot check our results, but, even so, these results allow us to speculate that there may indeed be semantic content in the manuscript. The discourse segmentation analysis also revealed interesting patterns, showing how several sections of the manuscript seem to have topical coherence reflected in their internal similarity scores while others, such as the Cosmological and Astronomical sections, seem to change sub-topics with each page turn. Languages A and B, on the other hand, are shown to display a surprising degree of *dissimilarity*, a result that directly contrasts with some research in this area but that supports other results [Lindemann and Bower 2021].

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The present collaboration between a computer scientist (Layfield) and a medieval codicologist (Davis) demonstrates the importance of interdisciplinary methodologies where the Voynich Manuscript is concerned. The manuscript is such a complex object that no one person could hope to master all of the fields necessary to undertake a thorough investigation of its contents, illustrations, structure, and history. Neither of the present authors could have reached these results without the other. Scientific and humanistic collaboration is crucial. It is only by reaching across the disciplinary divides that we will be able to make progress in understanding the mysteries of the Voynich Manuscript.

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The present study has investigated the coherence and flow of the Voynich Manuscript from a big-picture perspective; in forthcoming work (currently under review), the authors will present more granular results that consider how LSA can shed light on the original codicological structure of the manuscript. It is our hope that these results will assist others in their work on the Voynich Manuscript, moving us all closer to a deeper understanding of the World’s Most Mysterious Manuscript.

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Appendix 1: History and Methodology of LSA

Methodology

In order to generate our semantic space, we first need a set of m terms and n documents. The list of terms should be all the unique terms from the collection of text overall. The documents should be the set of documents (these can range from sentences and paragraphs to an entire large piece of text) from which we wish to create our semantic space. From

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this dataset we generate a m by n (*term by document*) matrix A . Each A_{ij} value in this matrix indicates the number of times term i occurs in document j , the so-called Term Frequency (TF).

The next step is to compute the Singular Value Decomposition (SVD) of the Matrix A . This is a form of factor analysis and factors our matrix A into three matrices:

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$$A = TSD^T$$

These matrices have the following shape:

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$$\begin{array}{c} \text{documents} \\ \left[\begin{array}{c} A \\ m \times n \end{array} \right] = \left[\begin{array}{c} T \\ m \times r \end{array} \right] \left[\begin{array}{c} S \\ r \times r \end{array} \right] \left[\begin{array}{c} D^T \\ r \times n \end{array} \right] \end{array}$$

terms

Figure 9. SVD Matrix Factorization

For a full mathematical description of the factors, the reader is referred to the original paper [Deerwester et al. 1990]. The T and D^T matrices will be almost entirely nonzero, and the S matrix is actually a diagonal matrix of the singular values from the computation in descending order.

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Roughly speaking, the D^T matrix refers to the documents from which we have built this space. Each column corresponds to the document in the column space of A (which, most importantly, is *non sparse*). The next step in this process is the dimensionality reduction phase. This involves picking a value, k , and reducing the dimensionality of our T , S and D^T matrices such that they are $m \times k$, $k \times k$, and $k \times n$ — that is we keep the first k columns in T , the first k values in the diagonal matrix S and the first k rows in D^T . This results in an approximation of A . More importantly, we reduce the dimensionality in order to filter out some of the noise in the space and focus on the key relationships found within. The selection of this value is somewhat subjective. In the work by [Foltz, Kintsch, and Landauer 1998], they use a value of 300 for k as their original A matrix is of a moderate size. There are some suggested methodologies for computing an appropriate value of k (such as choosing a k such that the ratio of the sum of the squares of the first k singular values divided by the sum of the entire set of squared singular values is about 0.8), but we have found that for smaller datasets they tend to work quite poorly. As the term-by-document matrices we will be dealing with (such as the text from the VM) we have used a value of 75 for k . Note the matrices we will be using are much smaller than the ones used in the original experiment in [Foltz, Kintsch, and Landauer 1998] so a smaller dimensionality is justified. We give a justification of this value next.

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The dimensional reduction step in LSA involves a selection of the number of dimensions to use from the SVD decomposition. This is a tricky choice, and much of the literature (especially in the context of Information Retrieval (IR)) suggests you pick a k that gives you the best results (for a typical train/test split for an IR problem this is straightforward). As we don't know the contents of the Voynich, this is clearly impossible, so we use three heuristics to approximate what might be a good value for k by calculating what might be a good lower, mid, and upper bound selection of values. This is actually done for all the dataset's semantic spaces. Note, these heuristics are designed for a semantic space generated from a *single text* which may not be typical in general. Given the results from [Foltz, Kintsch,

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and Landauer 1998], we know that sequential document comparisons should have a larger value than the shuffled variant as per our first experiment; so we first calculate a coherence score which is the average similarity of adjacent documents in the D^T matrix. Next, we shuffle the order of the documents in D^T and make the same calculation (random) and do this 10 times taking the average. This is like our coherence experiment except we are using this to help calibrate k . We then use three different techniques to approximate k .

1. Noise separation: Compare the baseline coherence for a dimension with the random baseline. Select the k with the largest difference (indicates strong semantic pattern). This would act as the lower bound for k as typically the difference is largest the smaller the dimension (assuming the text has any coherence). If the dimension selected is quite low this runs the risk of over-simplification and causing the model to lose essential topic/latent information.

2. Elbow detection: Find point of diminishing returns in the coherence scores. This is often indicated by the “elbow” of the coherence scores over the range of k explored. The similarity scores at this point begin to taper off. This is typically the default value of k as it should be larger than the Noise separation value and smaller than the Slope stabilization value.

3. Slope stabilization: Identify when the coherence values begin to flatten out. As the dimension increases usually the scores decrease. When the scores flatten out additional dimensions contribute very little. This is typically the upper bound for a reasonable k .

We ran these heuristics over a k range of [25, 200]. We did this for all datasets used, TF, TF-IDF and Log-Entropy weighting methods as well as document sizes of 20, 30 and 40. The range of values for the noise separation methods was [25, 30], for elbow detection it was [65, 90] and for the slope stabilization method it was [115, 160]. A reasonable range for k would seem to be 65-90 and as the average value for the elbow method, overall, was 75 so that was selected as the dimension to use for all the semantic spaces (anything in the range 65-90 should work reasonably well).

To compare two documents in the D^T matrix we need to generate two vectors to perform the cosine operation on. If we wanted to compare document i and j (corresponding to columns i and j) we would generate their vectors as follows:

$$\vec{v}_i = S_k D_{ki}^T$$

$$\vec{v}_j = S_k D_{kj}^T$$

Each vector is comprised of the product of the S matrix (singular values) and the appropriate column in the D^T matrix. The subscript k indicates we are using the first k dimensions.

Then apply the cosine operator as follows:

$$\cos(\vec{v}_i, \vec{v}_j) = \frac{\vec{v}_i \cdot \vec{v}_j}{\|\vec{v}_i\| \|\vec{v}_j\|}$$

We can also compare documents that do not already exist as columns in the document semantic space. This is achieved via a process known as “folding-in” a document which essentially transforms it into the same vector space as the documents that already exist in D^T . So given a new document, d , that we want to use for comparison purposes, we put it into the same format as a column in the original A matrix; that is, each dimension of d indicates how many times a particular term occurs in the document. We then fold it into the D^T matrix (essentially as an extra column) as follows:

$$\vec{d}^1 = d T_k S_k^{-1}$$

At this point, the new document vector $\vec{d}^1$ can be treated just as any other document vector in D^T for comparison purposes.

Building the Models for the Coherence Experiment

One question that has yet to be answered is: What do we consider a “document” in the context of generating our initial term-by-document matrices? For the original coherence experiments in [Foltz, Kintsch, and Landauer 1998], they used a document size of one sentence and, in the discourse segmentation part of their experiments, they used collections of 10 paragraphs as a document unit. With the VM we have no real concept of “sentence” as there appears to be no punctuation in the writing. Bearing this in mind (and wanting to treat all documents equally) we opted to experiment with document sizes of 20, 30 and 40 terms respectively. These are simply constructed by taking term sets of 20, 30 and 40 terms from the processed texts and treating those as the document partitions for creating the initial term-by-document matrix A .

We experiment by applying additional NLP techniques to the A matrix to try and lessen the effect of more common terms in the comparisons. There is potentially a problem in only utilizing TF values in the A matrix in that very common terms will have more influence in comparison operations than perhaps they should, purely due to their common appearance over the text. Removal of stop words partially resolves this issue, but there will still be a disproportionately high presence of some terms over others. In addition to creating the A matrix as described, which uses straight TF values, two weighting schemes are also employed to modify the values contained in A . The two schemes applied are Term Frequency-Inverse Document Frequency (TF-IDF) and Log-Entropy (LE).

Log-Entropy is defined in the equation below (in the context of matrix A). f_{ij} represents the TF of term i in document j and $p_{ij} = \frac{f_{ij}}{\sum_j f_{ij}}$ which equates to the fraction of occurrences term i has in document j over all occurrences of term i and n is the number of documents (columns) in A . Log-Entropy was the matrix weighting mechanism used in the experiments by [Foltz, Kintsch, and Landauer 1998]. The first logarithm decreases the effect of the differences in term frequency (to help smooth them out) whilst the entropy calculation will tend to give less weight to frequently occurring terms; additionally, it takes the distribution of the terms over the document set into account.

$$A_{ij} = \log(1 + f_{ij}) \left[1 + \left(\sum_j \frac{p_{ij} \log(p_{ij})}{\log(n)} \right) \right]$$

TF-IDF is defined in the equation below. The value df_i represents the document frequency of term i ; that is, the number of unique documents that term i occurs in. The logarithm value is the inverse document frequency, and it is high if the term is rare, thus emphasizing the value of rare terms and de-emphasizing the values of common terms.

$$A_{ij} = f_{ij} \log \left(\frac{n}{df_i} \right)$$

It is worth noting that each semantic space is trained on its own document and comparisons/experimentation is done within that semantic space. This can be viewed as a *circular* relationship or, perhaps better stated, self-contained distributional modelling (in the sense the training data is what we are working on) and can potentially viewed as problematic. Our experiments are *not* testing *generalization* (an issue with over-fitting); instead, we are testing internal structure. The goal of our work is to characterise internal coherence and discourse structure rather than consider predictive performance or any sense of generalization; that is, do the different semantic spaces behave in a similar fashion?

There is also some precedence of the usage of a generated semantic space and experimentation on the same space with the data it was trained on. [Dos Santos and Favero 2015] built a LSA semantic space from a set of student answers

to exam questions. They evaluated the performance of this on the same dataset (comparing human assigned scores to computer generated scores) and reported an 85% agreement with human scoring. [Altszyler et al. 2017] applied LSA using a semantic space made up of dream reports analysing usage patterns of words between dreams and waking life (the analysis making use of the same semantic space data). They also compared the results with Word2Vec deducing that with smaller corpus sizes Word2Vec becomes less effective compared to LSA. In addition, the paper [Foltz, Kintsch, and Landauer 1998], from which we are taking our methodology, both for coherence and discourse segmentation, literally uses a semantic space trained on a dataset and then performs discourse segmentation on this space using the same dataset (no external data was used for training) which is exactly what we are doing.

Generating Stop-Word Lists

In order to generate a stop word list, terms are sorted by their Term Frequency (TF) count in descending order. Using a window of size w , we calculate the average difference in TF between the first w terms in our list. We move this window by 1 word and calculate this difference again. This process is repeated until this difference falls below a threshold value of v . If x steps were required to reach this threshold value, the first $(w + x - 1)$ words in our list are selected as our stop words. In other words, we look at the average decrease in the TF in a moving window until the rate of change falls below some value v . We utilize a window of size 6 and a threshold of 3 to generate reasonable stop word lists for English, Medieval Italian and Latin (essentially the top most frequent words given the cutoff threshold defined above). The VM stop word list is calculated using the same process and parameters. The stop word lists and their term frequencies are shown in Figures 10 through 16:

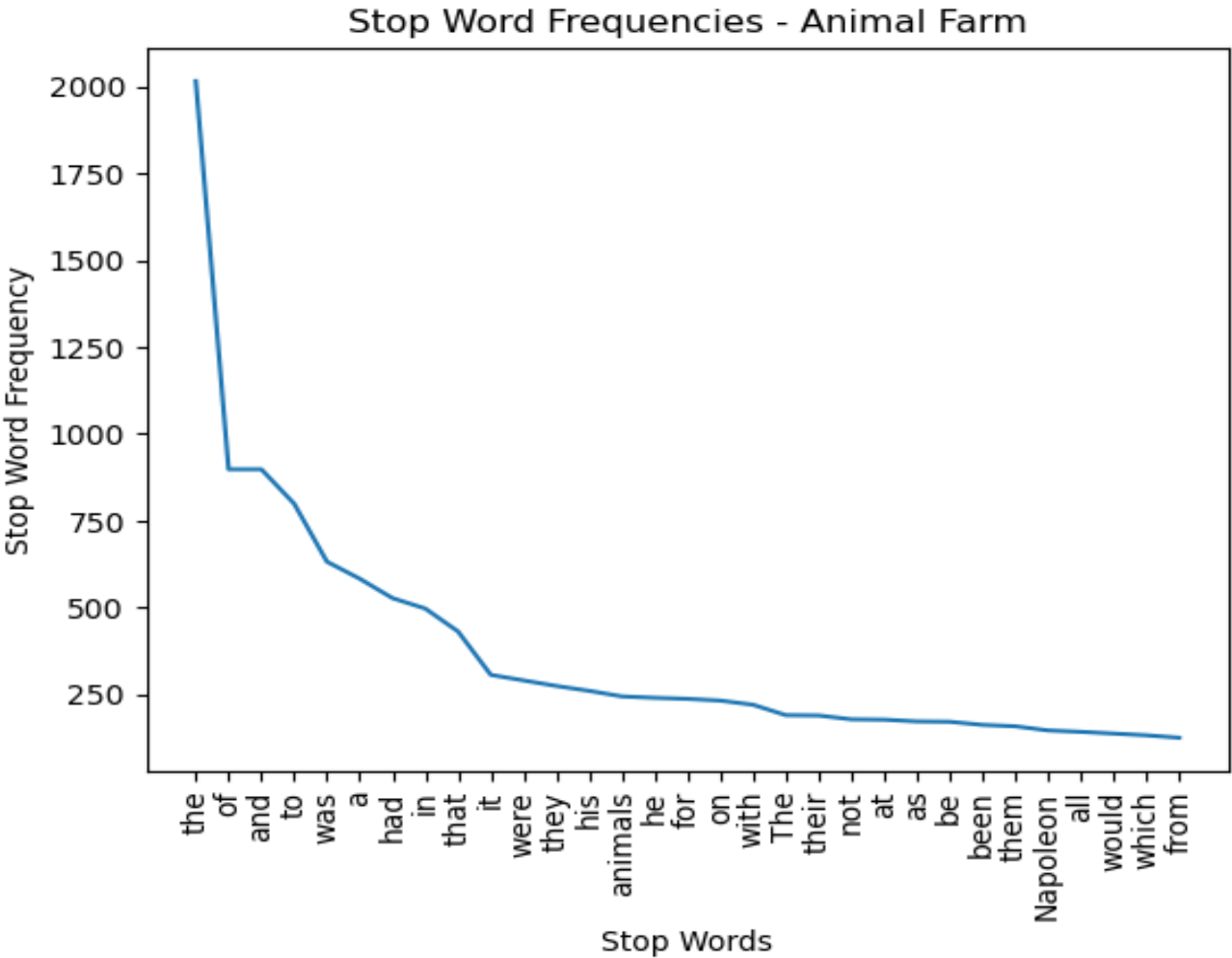


Figure 10. Stop Words – Animal Farm – English

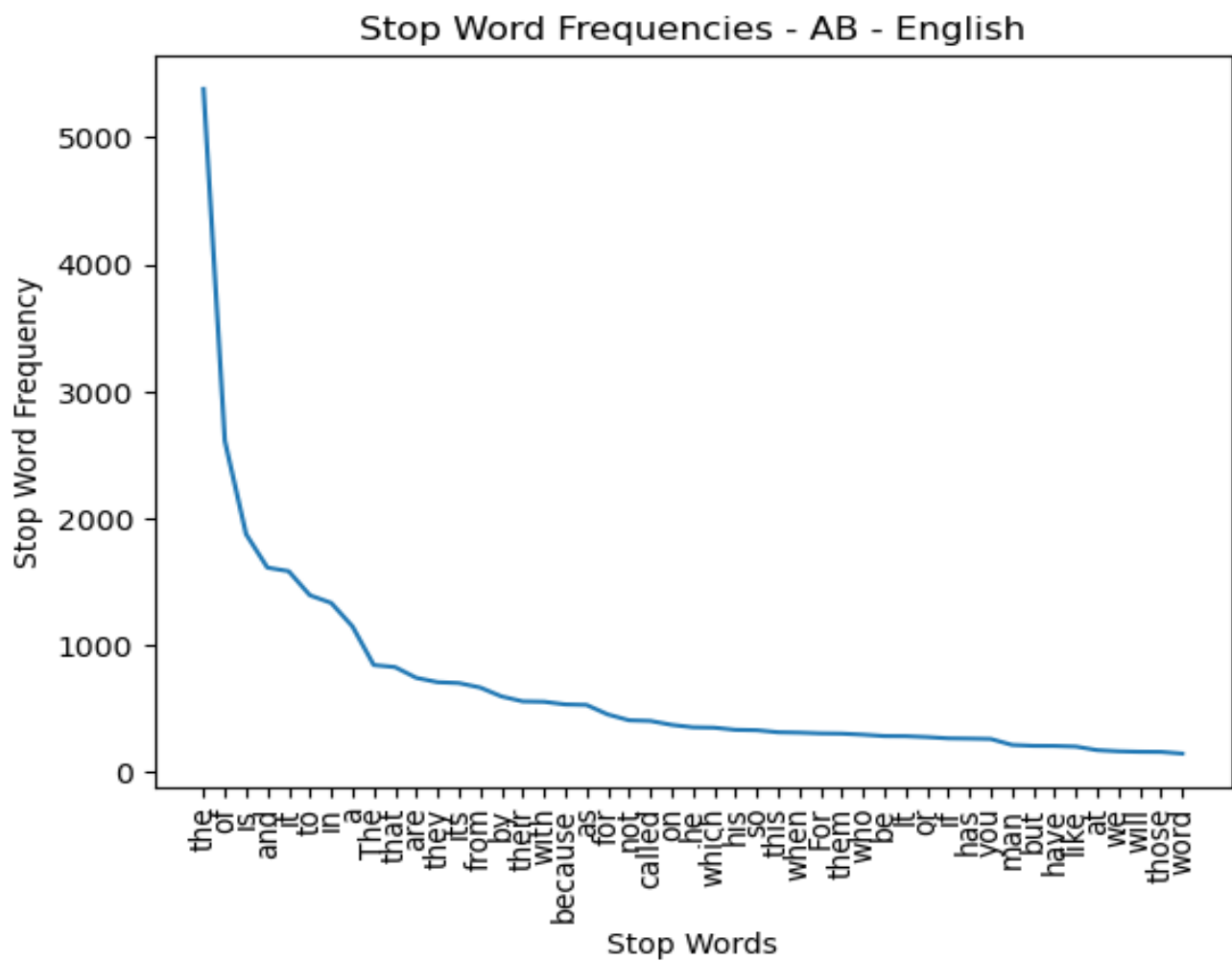


Figure 11. Stop Words – Aberdeen Bestiary – English

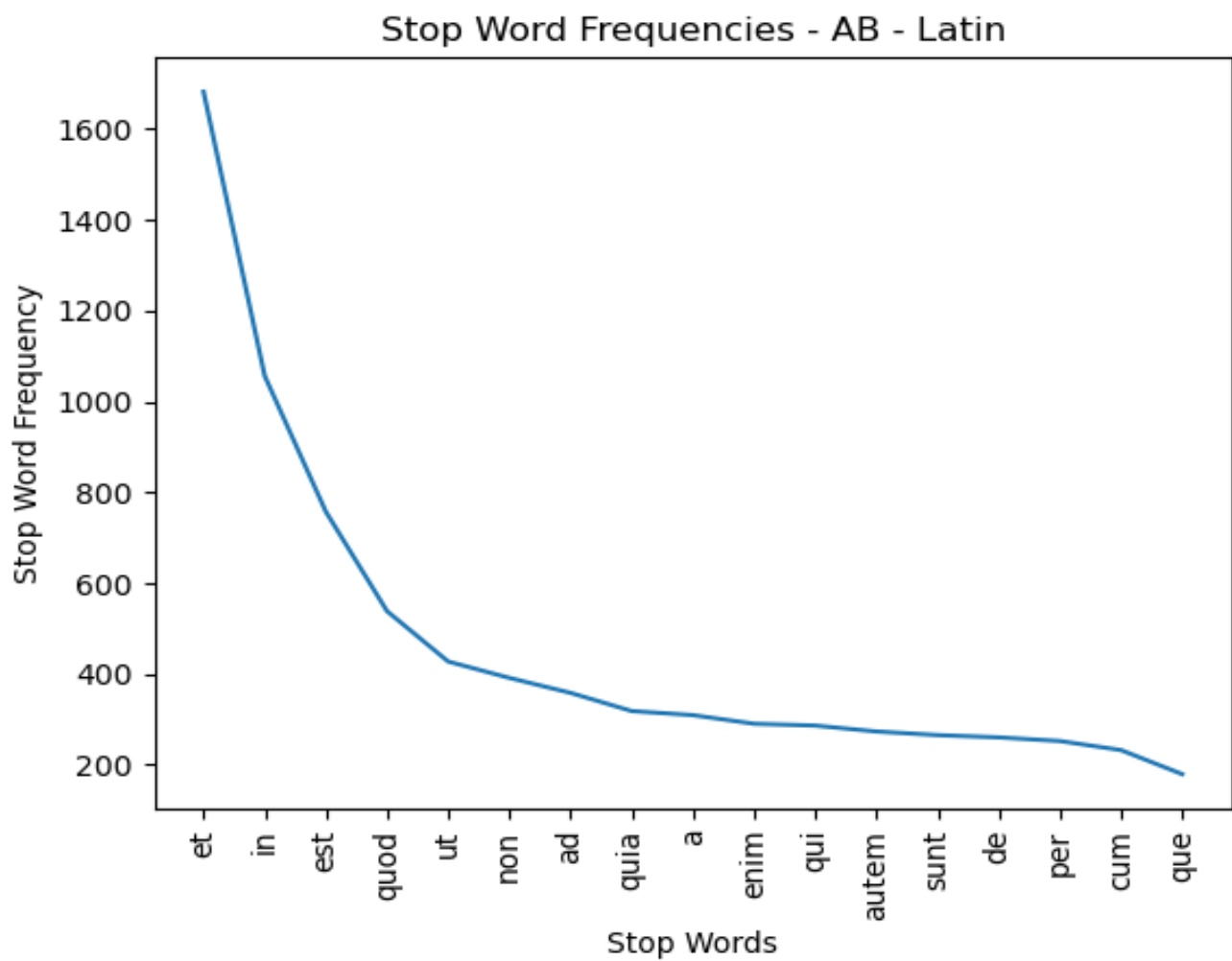


Figure 12. Stop Words - Aberdeen Bestiary – Latin

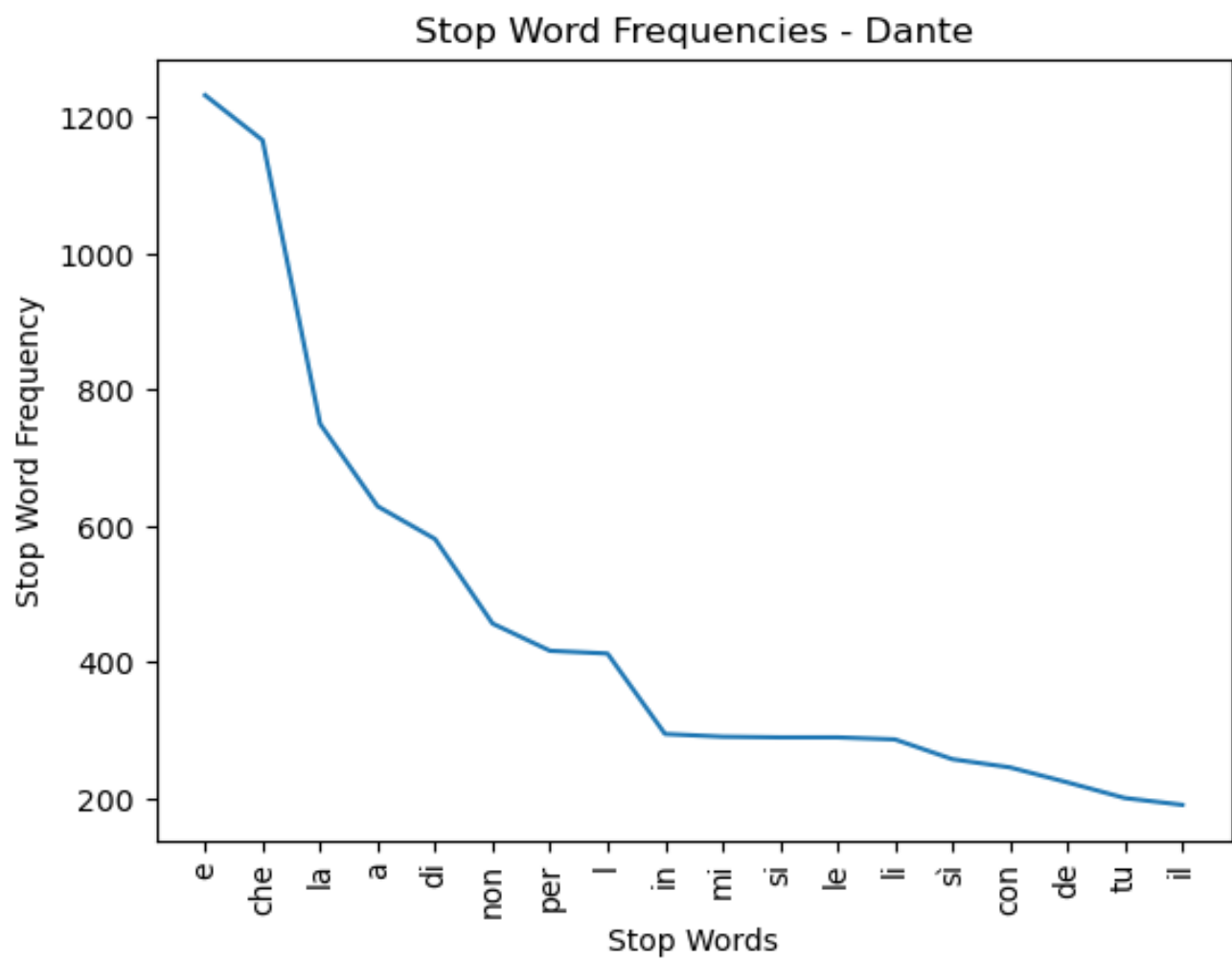


Figure 13. Stop Words – Dante's *Inferno* – Medieval Italian

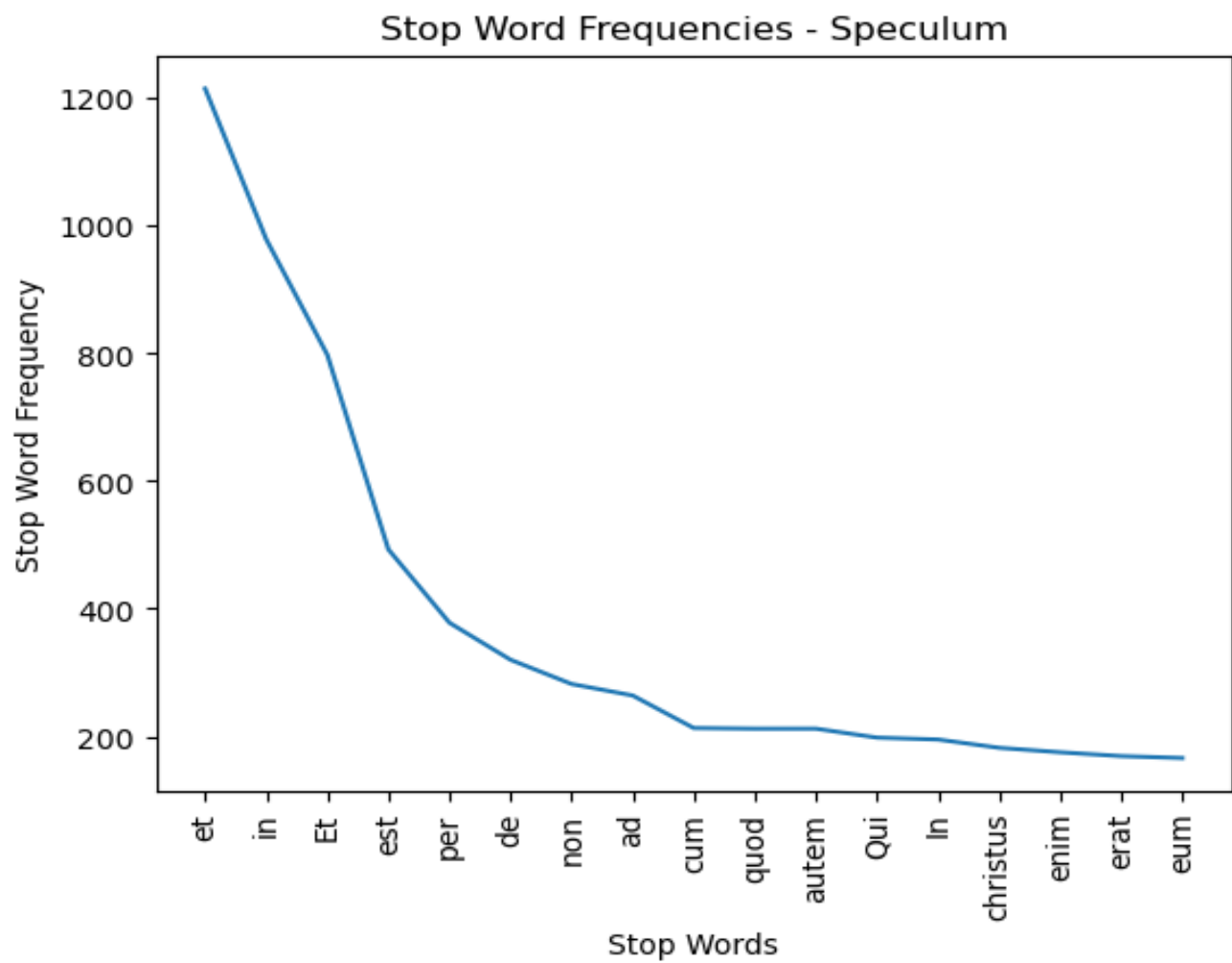


Figure 14. Stop Words – Speculum – Latin

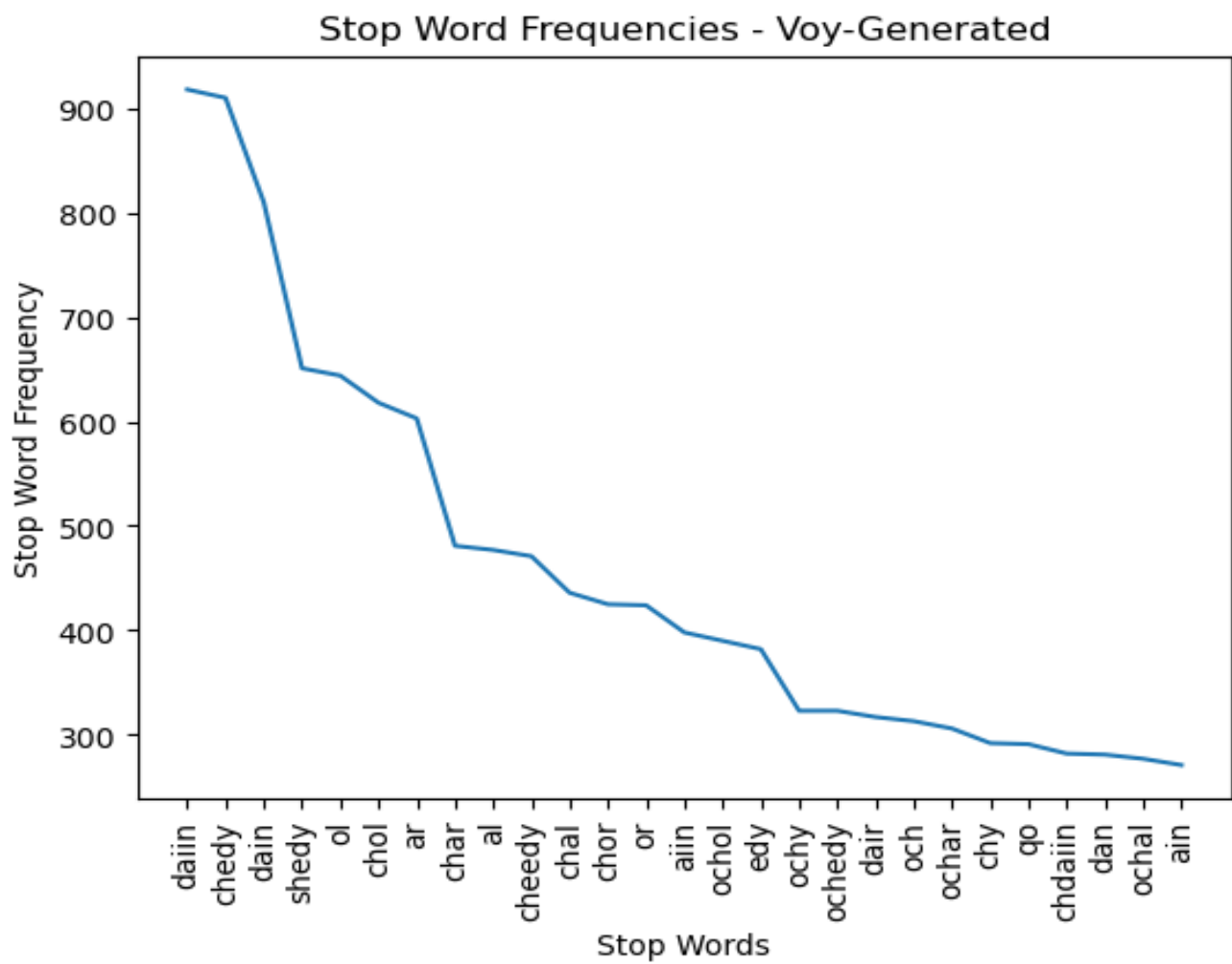


Figure 15. Stop Words – Generated Artificial Voynich

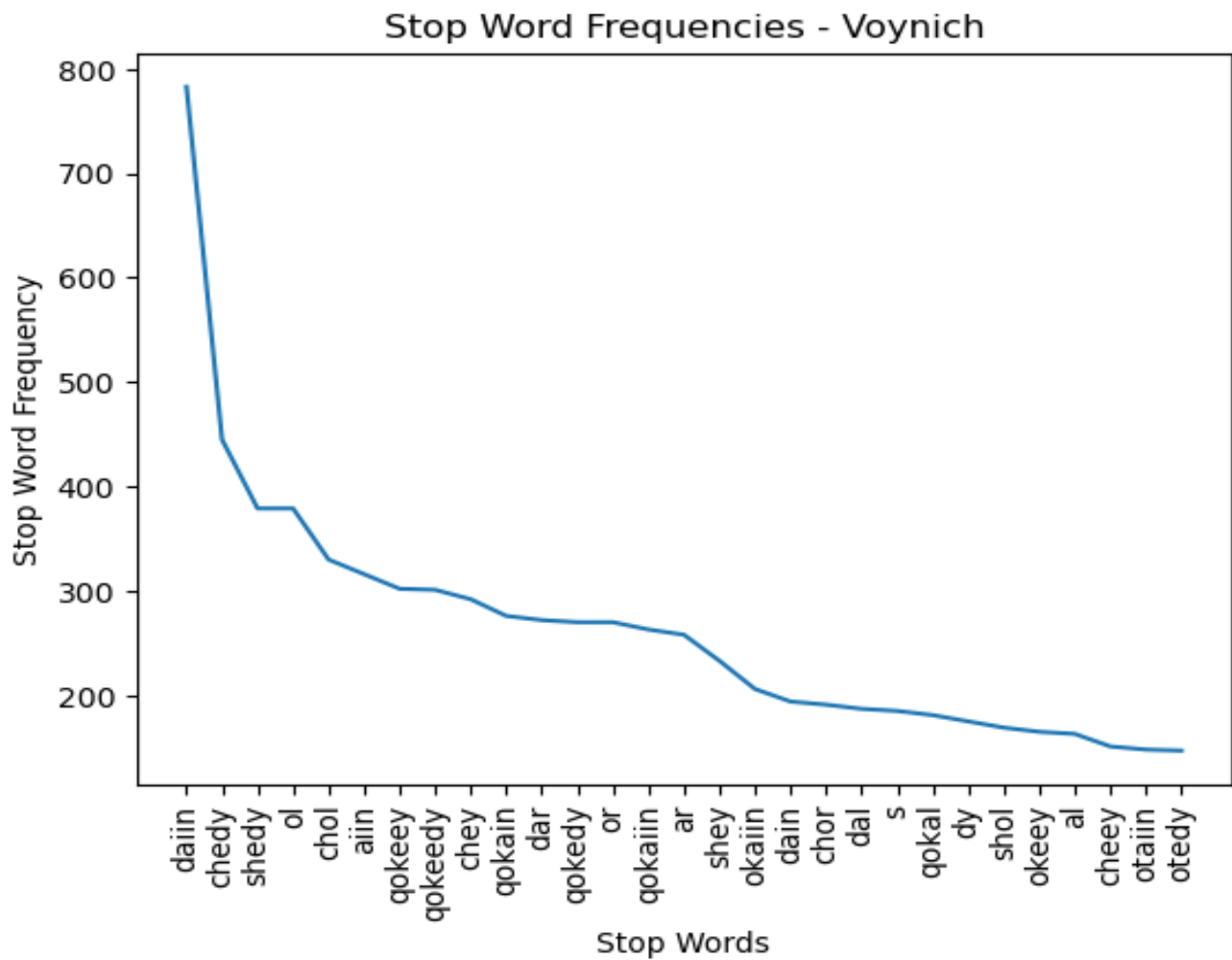


Figure 16. Figure 16: Stop Words for the Voynich Manuscript

Notes

[1] <https://beinecke.library.yale.edu/collections/highlights/voynich-manuscript>

[1] F. 116v is not part of the primary text and will not be considered here.

[2] The Appendix contains justification for the selected dimensional reduction as well as potential issues around how the semantic spaces themselves are constructed.

[3] Such as Term Frequency-Inverse Document Frequency (TF-IDF) or Log-Entropy. These are defined in Appendix 1.

[4] Again, TF-IDF and Log-Entropy are defined in Appendix 1.

[5] All the datasets used are described in Section 3.1

[6] The partition is split up into documents that, on average, are as close to 20 or 40 as possible as it's unlikely the text is exactly divisible by these values - text sections that are, in their entirety, less than half the target size, 20/40, are discarded.

[7] <https://www.abdn.ac.uk/bestiar/>

[8] The GitHub repository to the software can be found here: <https://github.com/TorstenTimm/SelfCitationTextgenerator/tree/master?tab=readme-ov-file>

[9] The text generated using Timm and Schinner's process used the default parameters except for "lines_to_create" where the value 3,500 was substituted (to generate more text) and the parameter "method.canFollow" which was set to the value "statistic" as this bases the generated text on the statistics for the VM.

[10] The transliteration files can be found here: <https://www.voynich.nu/transcr.html>

[11] The original [Foltz, Kintsch, and Landauer 1998] paper used a cutoff point of 2 standard deviations - as our results do not follow a normal distribution, this would have made little sense, so we chose a cutoff point of 5%.

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